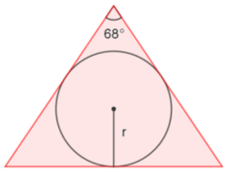
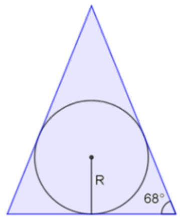
S2Q22

Question 22 - Stage 2



Area 68



Area 68

For any value of A with $60 < A < 90$ there are two isosceles triangles with area A square units and **largest** interior angle $A°$.

The diagram shows the two isosceles triangles in the case $A = 68$, along with their incircles.

For what value of A (where $60 < A < 90$) is the sum of the areas of the two incircles the greatest? Give your answer to 2 d.p.

Working

Triangles

let α be the unique angle of the triangle
let β be the other two
let l be the duplicate length of the triangle
let s be half of the shortest length

Red

$$\alpha = A\backslash\text{degree}$$
$$\beta = \frac{180-\alpha}{2}$$
$$A = \frac{1}{2}l^2\sin(\alpha)$$
$$l = \sqrt{\frac{2A}{\sin(\alpha)}}$$
$$s = l\cdot\sin\left(\frac{\alpha}{2}\right)$$
$$r = s\cdot\tan\left(\frac{\beta}{2}\right)$$
$$r = l\cdot\sin\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right)$$
$$r = \sqrt{\frac{2A}{\sin(\alpha)}}\cdot\sin\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right)$$
$$r = \sqrt{\frac{2A}{\sin(A)}}\cdot\sin\left(\frac{A}{2}\right)\cdot\tan\left(\frac{180-A}{4}\right)$$

Blue

$$\alpha = 180-2\beta$$
$$\beta = A\backslash\text{degree}$$
$$A = \frac{1}{2}l^2\sin(\alpha)$$
$$l = \sqrt{\frac{2A}{\sin(\alpha)}}$$
$$s = l\cdot\sin\left(\frac{\alpha}{2}\right)$$
$$R = s\cdot\tan\left(\frac{\beta}{2}\right)$$
$$R = l\cdot\sin\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right)$$
$$R = \sqrt{\frac{2A}{\sin(\alpha)}}\cdot\sin\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right)$$
$$R = \sqrt{\frac{2A}{\sin(180-2A)}}\cdot\sin\left(\frac{180-2A}{2}\right)\cdot\tan\left(\frac{A}{2}\right)$$

Circles

In [9]:
```python
from math import sin, tan, radians, pi

# Define degrees functions
def sinD(angle):
    return sin(radians(angle))

def tanD(angle):
    return tan(radians(angle))


lowerR = lambda A: ((( (2*A) / (sinD(A)) )**0.5) * sinD(A/2) * tanD((180-A)/(4)))
upperR = lambda A: ((( (2*A) / (sinD(180-2*A)) )**0.5) * sinD((180-2*A)/(2)) * tanD(A/2))
area = lambda r: pi*r**2

maximum = -1
maximumA = -1
a = 60
```

```python
while a < 90:
    a += 0.01 # Increment equal to the precisian
    current = area(lowerR(a)) + area(upperR(a))
    if isinstance(current,float) and current > maximum:
        maximum = current
        maximumA = a

print(round(maximumA,2)) # Round to deal with floating point imprecision
```

```
72.81
```